Computational Neuroscience (EC60007)
Project II

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1 Morris–Lecar model

Q1. Consistent units

(a) t = milliseconds, $I = \mu\text{A/cm}^2$, V = mV, $C = \mu\text{F/cm}^2$, and $g = \text{mS/cm}^2$.

Since

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = I,$$

we have

$$C := \frac{\mu\text{F}}{\text{cm}^2}, \qquad \frac{\mathrm{d}V}{\mathrm{d}t} := \frac{\text{mV}}{\text{ms}}.$$

Therefore

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mu\text{F}}{\text{cm}^2}\frac{\text{mV}}{\text{ms}} = \frac{\mu\text{A}}{\text{cm}^2}.$$

Also, $F = C_{\text{charge}}/V$, so $FV = C_{\text{charge}}$. Hence

$$\mu\text{F} \times \text{mV} = 10^{-6} \times 10^{-3}\,\text{F V} = 10^{-9}\,\text{C},$$

and

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{10^{-9}\,\text{C}}{\text{cm}^2\,10^{-3}\,\text{s}} = \frac{10^{-6}\,\text{C}}{\text{cm}^2\,\text{s}} = \frac{\mu\text{A}}{\text{cm}^2}.$$

Thus these units are consistent.

For the conductance term,

$$I = g(V-E) = \frac{\text{mS}}{\text{cm}^2}\,\text{mV} = \frac{10^{-6}\,\text{S V}}{\text{cm}^2} = \frac{\mu\text{A}}{\text{cm}^2},$$

which is also consistent.

(b) If $g := \mu\text{S/cm}^2$ and we take $V = \text{mV}$, then

$$I = gV = \frac{\mu\text{S}}{\text{cm}^2}\,\text{mV} = \frac{10^{-9}\,\text{S V}}{\text{cm}^2} = \frac{\text{nA}}{\text{cm}^2}.$$

So $I := \text{nA/cm}^2$. The solution is not unique and depends on the chosen unit of voltage.

Q2. Equilibrium for $I_{\text{ext}} = 0$

The Morris–Lecar equations are

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = I_{\text{ext}} - \bar{g}_{\text{Ca}} m_\infty(V)(V - E_{\text{Ca}}) - \bar{g}_{\text{K}} w (V - E_{\text{K}}) - \bar{g}_{\text{L}}(V - E_{\text{L}}), \tag{1}$$

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \phi\frac{w_\infty(V) - w}{\tau_w(V)}, \tag{2}$$

where

$$m_\infty(V) = \frac{1}{2}\left[1 + \tanh\left(\frac{V - V_1}{V_2}\right)\right], \tag{3}$$

$$w_\infty(V) = \frac{1}{2}\left[1 + \tanh\left(\frac{V - V_3}{V_4}\right)\right], \tag{4}$$

$$\tau_w(V) = \frac{1}{\cosh\left((V - V_5)/V_6\right)}. \tag{5}$$

For equilibrium:

M-1: $\mathrm{d}V/\mathrm{d}t = 0$ and $\mathrm{d}w/\mathrm{d}t = 0$. For $I_{\text{ext}} = 0$, the voltage equation gives

$$w = \frac{-\bar{g}_{\text{Ca}} m_\infty(V)(V - E_{\text{Ca}}) - \bar{g}_{\text{L}}(V - E_{\text{L}})}{\bar{g}_{\text{K}}(V - E_{\text{K}})}. \tag{i}$$

The recovery equation gives

$$w = w_\infty(V) = \frac{1}{2}\left[1 + \tanh\left(\frac{V - V_3}{V_4}\right)\right]. \tag{ii}$$

Equations (i) and (ii) were combined into one variable equation and solved using `fzero`.

M-2: We solved $\dot{V} = 0$ and $\dot{w} = 0$ simultaneously using `fsolve`. Both methods give

$$V^* \approx -60.855382\ \text{mV}, \qquad w^* \approx 0.01491502.$$

Q3. Stability at rest

By looking at the arrows in the quiver plot, we can say that it is a stable equilibrium. The Jacobian and

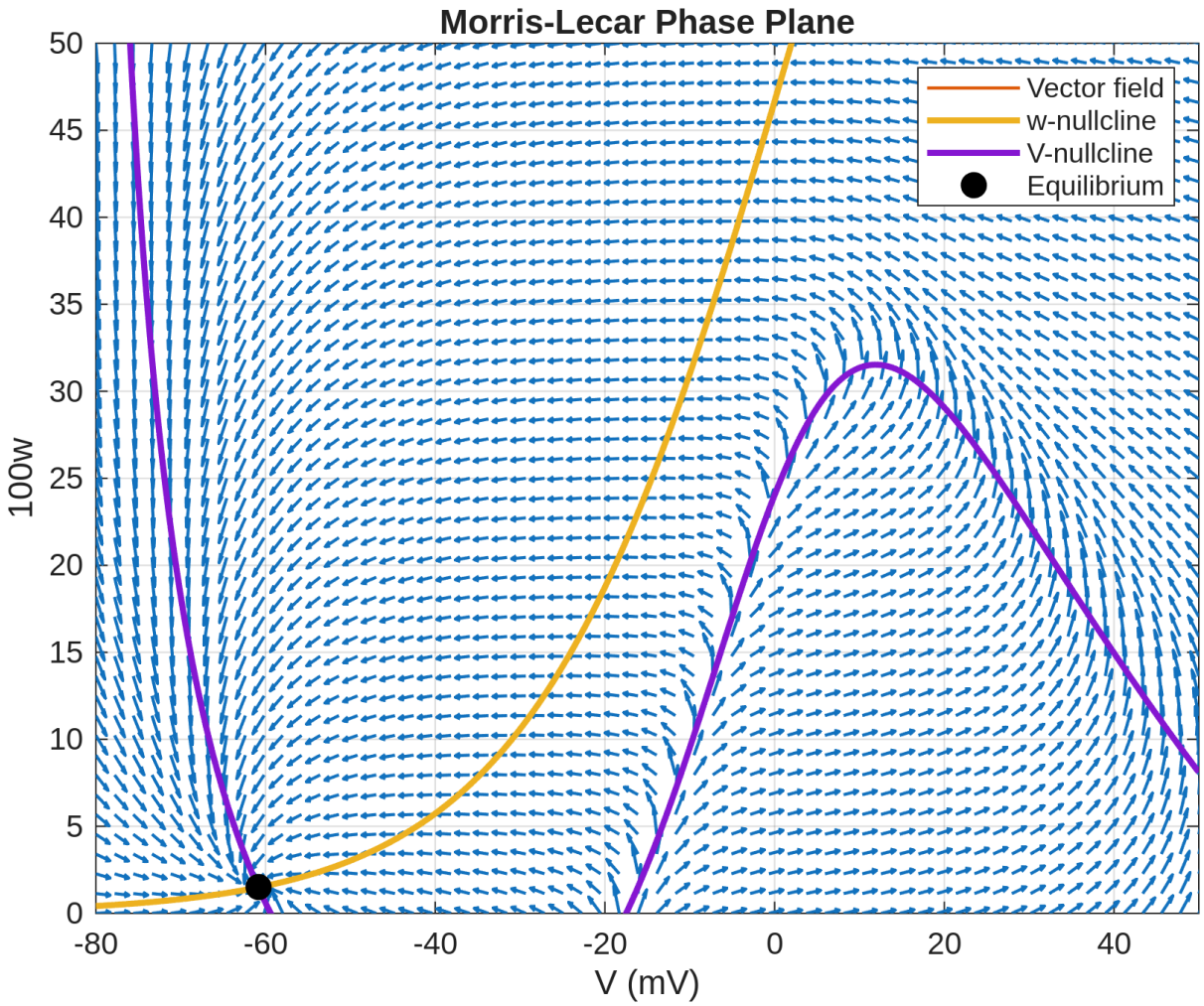


Figure 1: Q2: Morris–Lecar phase plane at $I_{\text{ext}} = 0$. The voltage and recovery nullclines intersect at the equilibrium (V^*, w^*), while the arrows show the local flow. The vertical axis is plotted as $100w$ for visibility.

eigenvalues are

$$J^* = \begin{pmatrix} -0.1004 & -9.2578 \\ 0.0001 & -0.0825 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_{1,2} = -0.0915 \pm 0.0258i.$$

Therefore $\text{Re}(\lambda) < 0$. The trajectory spirals inward, so the equilibrium is a stable spiral.

Q4. Numerical tolerances

For MLE, we are dealing with values around $V \sim 10$–100 mV and $w \sim 10^{-2}$–1. So RelTol $= 10^{-3}$ is small, since it is a relative tolerance of 0.1%. AbsTol $= 10^{-6}$ is also small on the absolute scale. Therefore these tolerances work for our selected unit setup.

If we change mV to kV, RelTol remains the same because it is relative. But AbsTol should be changed because

$$10^{-6} \text{ mV} = 10^{-12} \text{ kV}.$$

So the voltage absolute tolerance becomes 10^{-12} in kV. The tolerance for the dimensionless variable w remains unchanged.

Q5. Effect of ϕ on an action potential

Since

$$\dot{w} = \phi \frac{w_\infty(V) - w}{\tau_w(V)},$$

increasing ϕ increases the rate of change of w. Since w controls the potassium current, when w changes rapidly there is less time for V to increase or stay at a particular value. Hence, in the action potential (V versus t), as ϕ increases the amplitude of V decreases and the voltage returns to equilibrium quickly. Also, the nullclines are independent of ϕ.

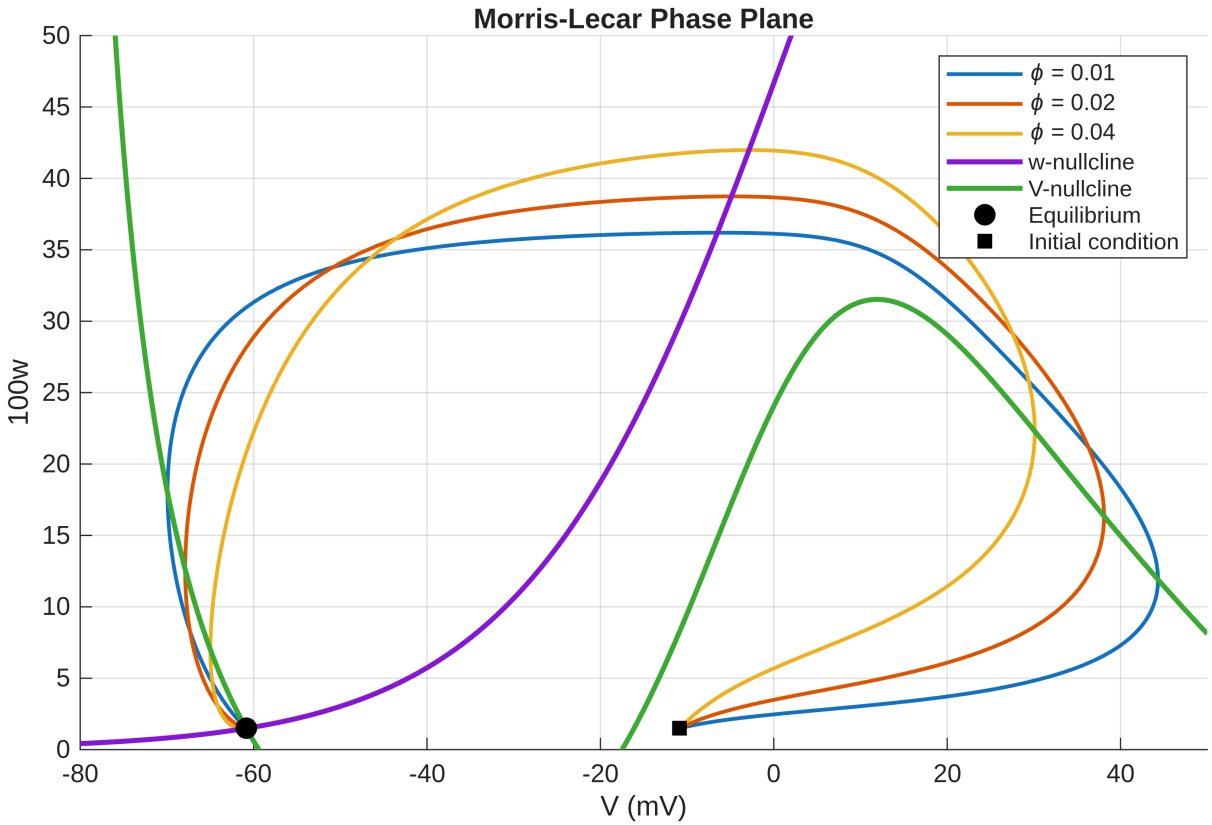


Figure 3: Q5: Phase-plane trajectories for the three values of ϕ. The nullclines and equilibrium do not depend on ϕ, but the trajectories change because the relative speed of the recovery variable changes. The vertical axis is $100w$.

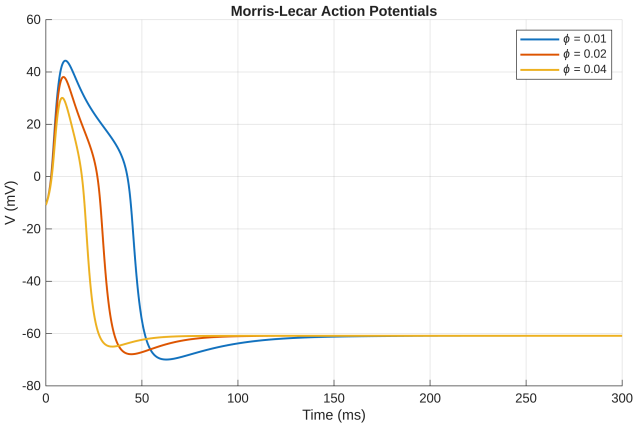


Figure 2: Q5: Morris–Lecar action-potential waveforms for $\phi = 0.01$, 0.02, and 0.04. Increasing ϕ accelerates recovery, reduces the peak, and shortens the voltage excursion.

Q6. Threshold depolarization

For $I_{\text{ext}} = 0$, $w(0) = w^*$. As we varied the initial voltage, we noticed a sharp transition near a 46 mV depolarization. We then repeatedly divided the range and found the threshold to be

$$V_{\text{th}} = V^* + 45.990970 = -14.864412 \text{ mV}.$$

Required depolarization is approximately 45.99 mV. Yes, the MLE with the given initial conditions and parameters has threshold depolarization behaviour. Here threshold means the smallest initial increase in membrane potential above the resting equilibrium, with w initially fixed to w^*, that produces an action potential.

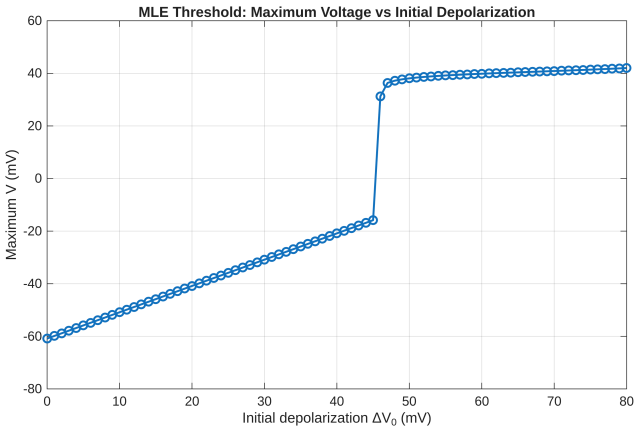


Figure 5: Q6: Maximum voltage reached as a function of initial depolarization. The abrupt jump near a 46 mV depolarization identifies the transition from a subthreshold response to a full action potential.

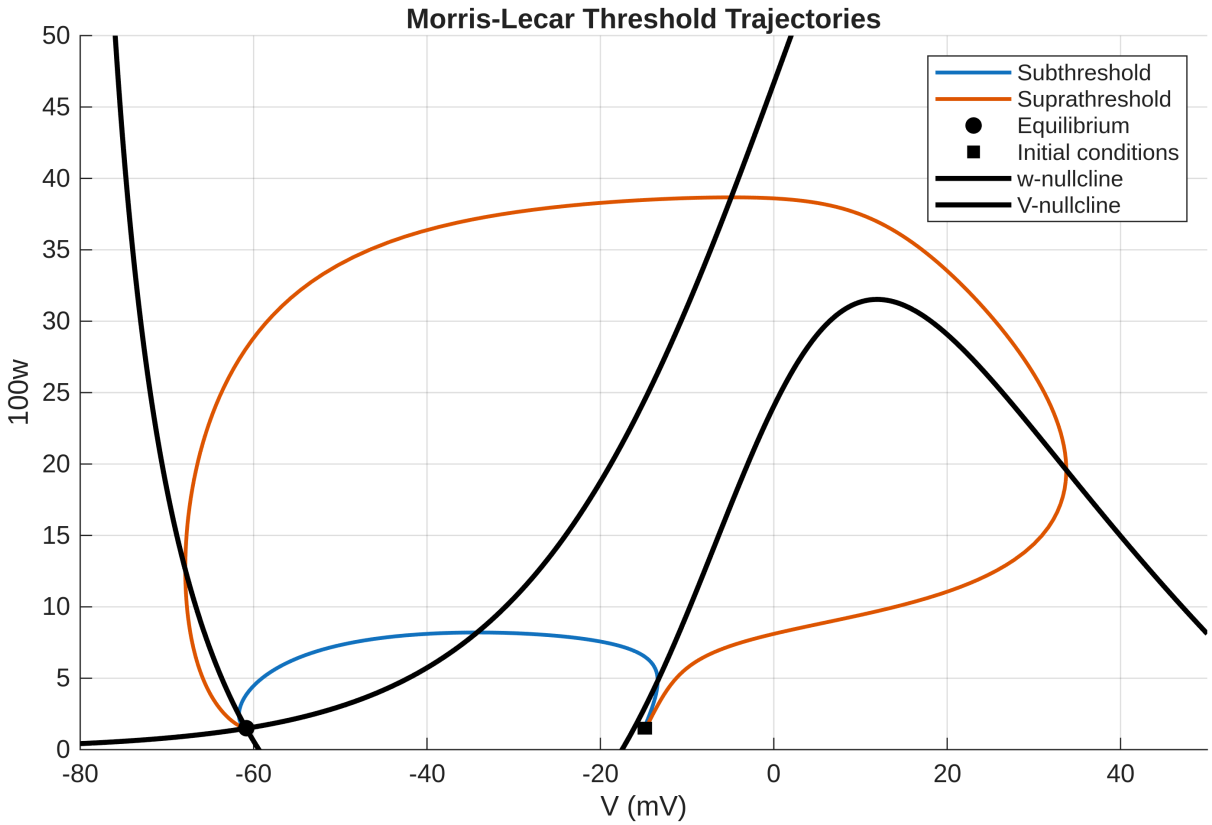


Figure 4: Q6: Representative phase-plane trajectories immediately below and above threshold. The subthreshold trajectory returns to the stable equilibrium, whereas the suprathreshold trajectory follows the large action-potential excursion. The vertical axis is $100w$.

Q7. Bistability at $I_{\text{ext}} = 86\ \mu\text{A/cm}^2$

For $I_{\text{ext}} = 86\ \mu\text{A/cm}^2$, we recalculated the equilibrium and found

$$(V^*, w^*) = (-27.952413\ \text{mV},\ 0.11953641),$$
$$\lambda_{1,2} = -0.0109 \pm 0.0667i.$$

Since $\text{Re}(\lambda) < 0$, it is a stable inward spiral.

In T-1, the trajectory approaches a stable limit cycle. In T-2, the trajectory remains as it is because it starts at the actual equilibrium. In T-3, the trajectory converges toward the calculated equilibrium point.

In T-1, we start at the equilibrium corresponding to $I_{\text{ext}} = 0$. Then $I_{\text{ext}} = 86\ \mu\text{A/cm}^2$ is applied, which causes the trajectory to enter the oscillating state. In T-2, we start at the equilibrium corresponding to $I_{\text{ext}} = 86\ \mu\text{A/cm}^2$. Applying the same current causes no change because the system is already at equilibrium.

Q8. Optional extra credit

This optional question was not attempted.

Q9. Equilibria and firing for $I_{\text{ext}} = 80$, 86, and 90

I_{ext} (μA/cm^2)	V^* (mV)	w^*	Eigenvalues
80	-29.966175	0.10611266	$-0.0226 \pm 0.0659i$
86	-27.952413	0.11953641	$-0.0109 \pm 0.0667i$
90	-26.596867	0.12937932	$-0.0020 \pm 0.0661i$

As I_{ext} increases, V^* shifts toward more positive values and the real part of the eigenvalues moves toward zero. This means that stability decreases slightly. The firing rate remains zero while the equilibrium is stable and becomes nonzero when it loses stability. The firing rate increases as the current is increased and is approximately 9.6 Hz at $I_{\text{ext}} = 93\ \mu\text{A/cm}^2$, as shown in the numerical firing-rate plot.

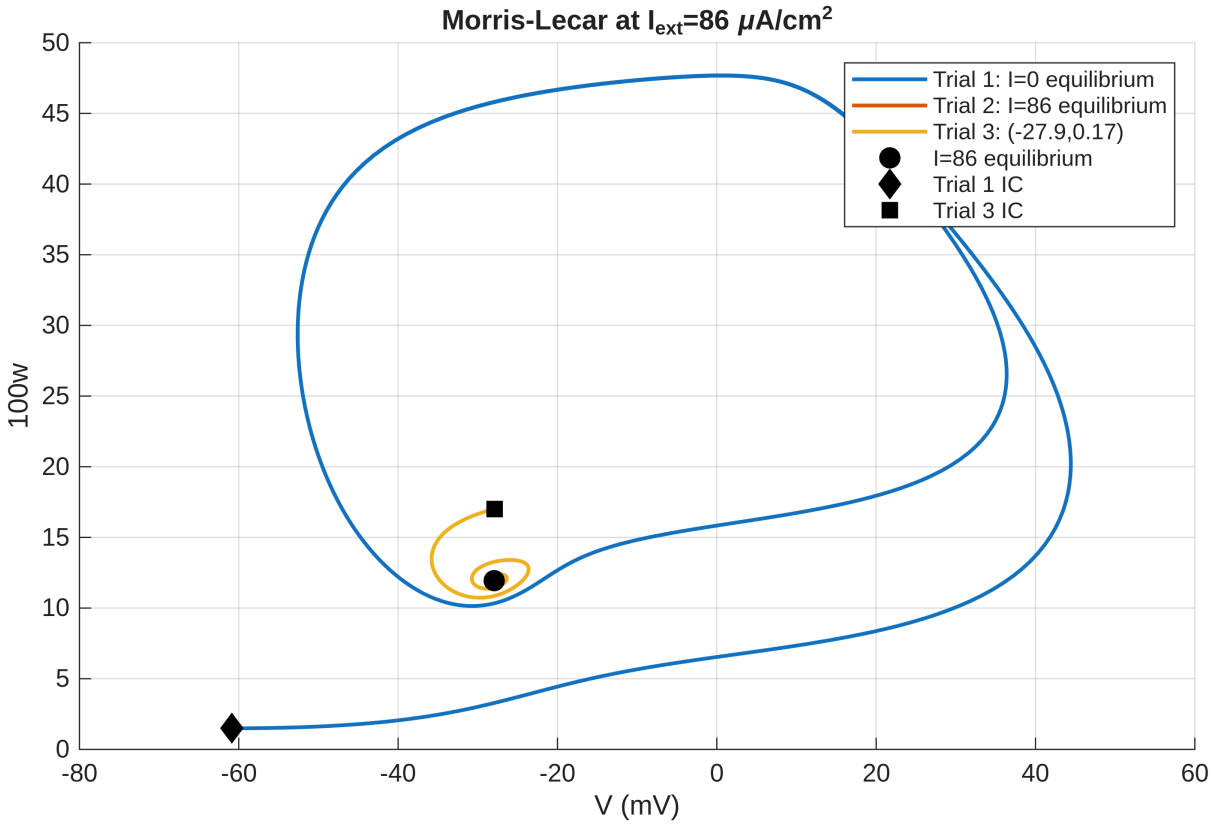


Figure 6: Q7: Morris–Lecar trajectories at $I_{\text{ext}} = 86\ \mu\text{A/cm}^2$. Starting from the zero-current equilibrium approaches the stable limit cycle; starting at the new equilibrium remains there; and the specified off-equilibrium initial condition spirals into the stable focus. The vertical axis is $100w$.

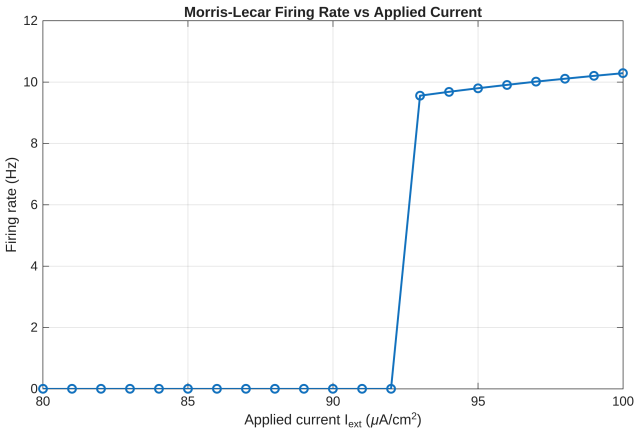


Figure 7: Q9: Morris–Lecar firing rate versus applied current. The rate is zero through 92 $\mu\text{A/cm}^2$, becomes nonzero at the onset of sustained firing, and then increases gradually with current.

Q10. Three equilibria at $I_{\text{ext}} = 30\ \mu\text{A/cm}^2$

For the second MLE parameter set, the equilibrium points are

	(V^*, w^*)	Eigenvalues	Classification
E_1	$(-41.845162,\ 0.00204747)$	$-0.0689,\ -0.7334$	stable equilibrium
E_2	$(-19.563243,\ 0.02588265)$	$0.1372,\ -0.1648$	saddle
E_3	$(3.871510,\ 0.28205130)$	$0.0911 \pm 0.1826i$	unstable spiral

For the saddle point, the unstable and stable eigenvectors are approximately

$$(0.999998,\ 0.001753)^T \quad \text{and} \quad (0.999909,\ 0.013470)^T,$$

respectively, and they define the corresponding manifolds. The unstable manifold leaves the saddle and one branch tends toward the stable equilibrium. The stable manifold converges to the saddle point.

Q11. Variation over 30–50 $\mu\text{A/cm}^2$

As I_{ext} is increased from 30 to 50 $\mu\text{A/cm}^2$, the saddle and stable equilibrium move toward each other. Near $I_{\text{ext}} \approx 39.96\ \mu\text{A/cm}^2$, these two collide and disappear. For $I_{\text{ext}} > I_{\text{collide}}$, only one equilibrium remains, and it is unstable.

Before the collision, the firing rate is zero because of the stable equilibrium. After the collision, the firing rate becomes nonzero and increases with I_{ext}.

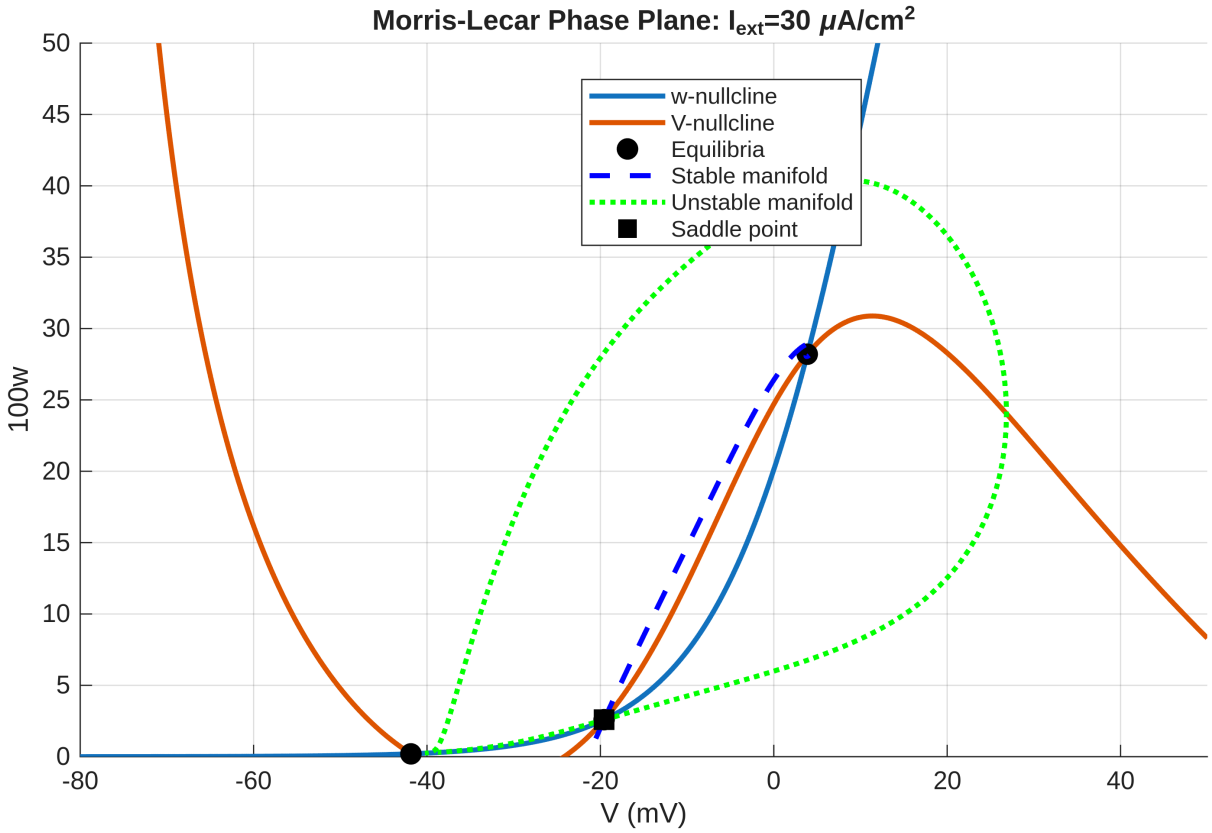


Figure 8: Q10: Morris–Lecar phase plane at $I_{\text{ext}} = 30\ \mu\text{A/cm}^2$. The plot shows both nullclines, all three equilibria, and the stable and unstable manifolds of the saddle point. The vertical axis is $100w$.

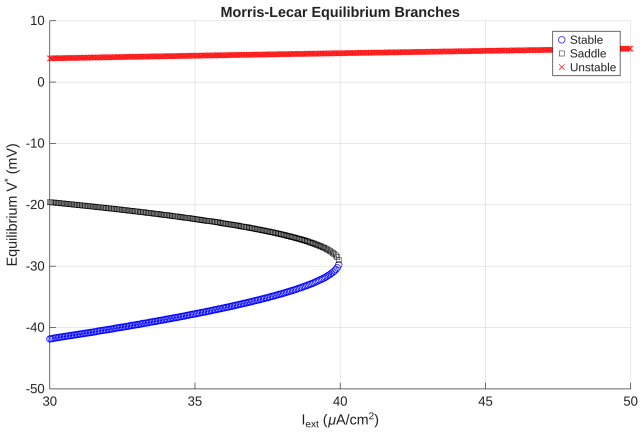


Figure 9: Q11: Morris–Lecar equilibrium-voltage branches over 30–50 $\mu\text{A/cm}^2$. The low-voltage stable branch and the saddle branch meet near 40 $\mu\text{A/cm}^2$, while the high-voltage equilibrium remains unstable.

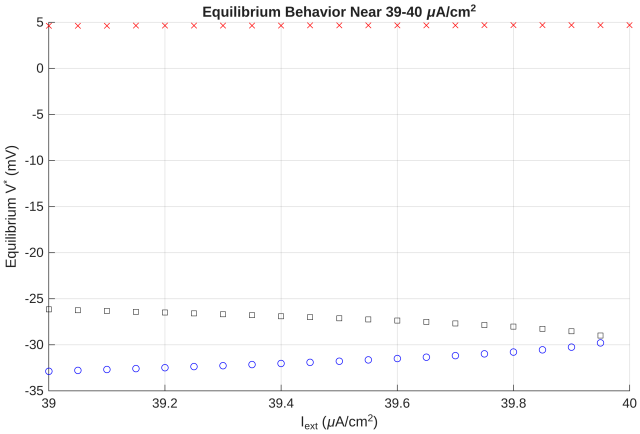


Figure 10: Q11: Enlarged view of the equilibrium branches between 39 and 40 $\mu\text{A/cm}^2$. The stable and saddle equilibria approach one another and disappear at the saddle-node bifurcation.

2 Hodgkin–Huxley model

Q12. Four-variable model

In this part, we successfully simulated the four-variable Hodgkin–Huxley equations with initial voltage $V_0 = -60$ mV and

$$x_0 = \frac{\alpha_x(V_0)}{\alpha_x(V_0) + \beta_x(V_0)}, \qquad x = n, m, h. \tag{6}$$

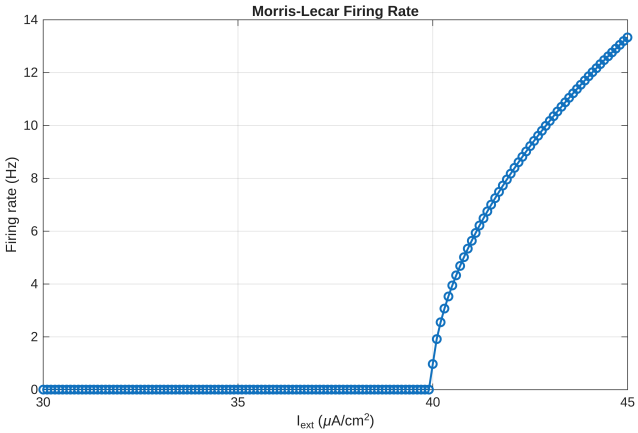


Figure 11: Q11: Morris–Lecar firing rate versus applied current. Firing begins near 40 μA/cm^2 and increases continuously as the current is raised to 45 μA/cm^2.

The voltage and gating equations are

$$C\dot{V} = I_{\text{ext}} - I_{\text{K}} - I_{\text{Na}} - I_{\text{L}}, \tag{7}$$
$$I_{\text{K}} = \bar{g}_{\text{K}} n^4 (V - E_{\text{K}}), \tag{8}$$
$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}}), \tag{9}$$
$$I_{\text{L}} = \bar{g}_{\text{L}} (V - E_{\text{L}}), \tag{10}$$
$$\dot{x} = \alpha_x(V)(1 - x) - \beta_x(V)x, \qquad x = n, m, h. \tag{11}$$

To handle the 0/0 cases for α_n and α_m, we calculated their limiting values:

$$\alpha_n(-50 \text{ mV}) = 0.1, \qquad \alpha_m(-35 \text{ mV}) = 1.$$

The temperature is 6.3°C, so $\phi = 1$.

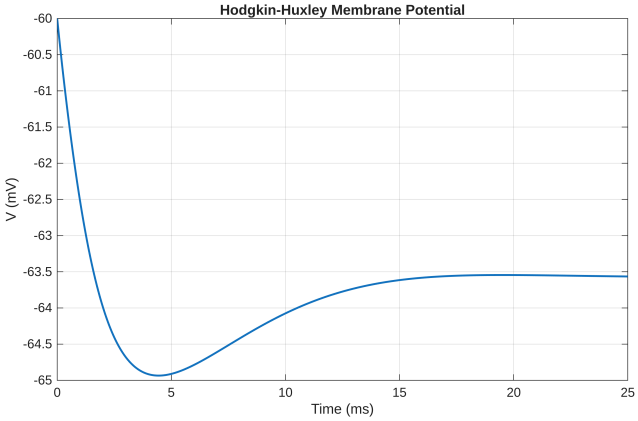


Figure 12: Q12: Membrane potential in the initial full HH simulation using the provisional leak reversal potential. The voltage relaxes from −60 mV toward approximately −63.56 mV, showing that E_{L} must be adjusted to obtain the required resting voltage.

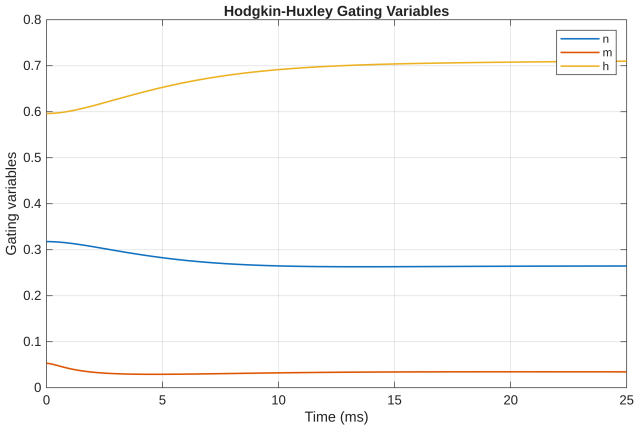


Figure 13: Q12: Evolution of the gating variables n, m, and h during the initial full HH simulation. The variables approach their steady values as the membrane relaxes.

Q13. Leak reversal potential

The membrane equation is

$$C\dot{V} = I_{\text{ext}} - I_{\text{K}} - I_{\text{Na}} - I_{\text{L}}.$$

At $V = -60$ mV, i.e. at equilibrium, $\dot{V} = 0$, $\dot{x} = 0$, and $I_{\text{ext}} = 0$. Therefore

$$I_{\text{L}} = -I_{\text{K}} - I_{\text{Na}}. \tag{i}$$

Also,

$$x(V) = \frac{\alpha_x(V)}{\alpha_x(V) + \beta_x(V)}, \qquad x = n, m, h. \tag{ii}$$

Calculating all the gating variables using (ii) gives

$$n_0 = 0.317677, \qquad m_0 = 0.052933,$$
$$h_0 = 0.596121.$$

Putting these values into (i), we get the leak reversal potential

$$E_{\text{L}} \approx -49.401079 \text{ mV}.$$

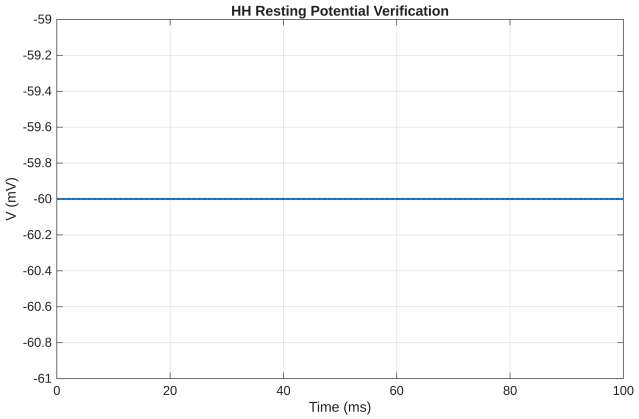


Figure 14: Q13: Verification of the calibrated HH resting state. With $E_{\text{L}} \approx -49.4011$ mV and $I_{\text{ext}} = 0$, the membrane remains at −60 mV throughout the simulation.

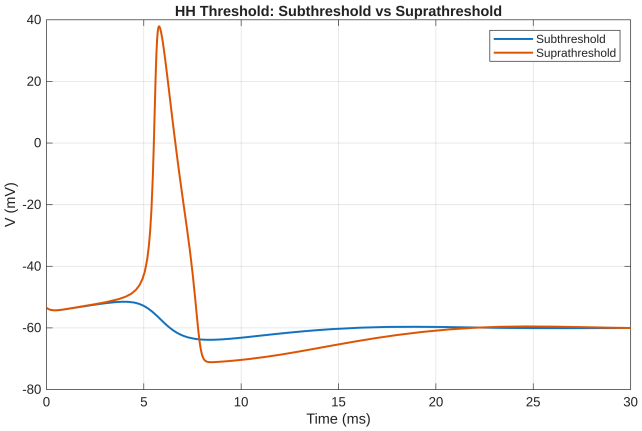


Figure 15: Q14: HH voltage responses immediately below and above the brief-depolarization threshold. The subthreshold response returns to rest, whereas the suprathreshold response produces a full action potential.

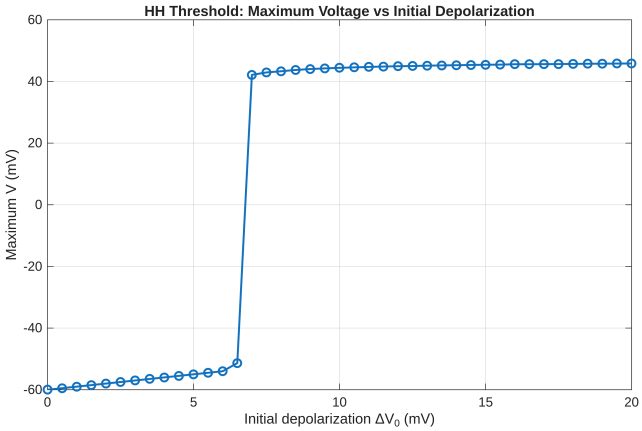


Figure 16: Q14: Maximum HH voltage versus initial depolarization. The sharp transition near $\Delta V = 6.51$ mV identifies the numerical threshold.

Q14. Resting stability and brief-pulse threshold

At rest, $V_R = -60$ mV. The eigenvalues are

$$-4.6753, \qquad -0.2027 \pm 0.3831i, \qquad -0.1207.$$

All real parts are negative; hence the resting state is stable.

For threshold, a brief current pulse is approximated by starting the system from $V_{\text{rest}} + \Delta V$. Depolarization below threshold causes a subthreshold response that returns to rest, whereas depolarization above threshold causes an action potential. We obtain

$$V_{\text{th}} \approx -53.492701 \text{ mV}, \qquad \Delta V \approx 6.507299 \text{ mV}.$$

The second initial point does not move because it is already at equilibrium. For the first initial point, the state is outside the basin of attraction of the stable equilibrium and inside the basin of the limit cycle. Thus a stable equilibrium and a stable limit cycle coexist.

Q15. Steady current injection from 8 to 12 μA/cm^2

A steady external current was applied from 8 to 12 μA/cm^2. For each individual case, the equilibrium was calculated. As current was increased, the equilibrium membrane potential increased almost linearly, and so did the real part of the Jacobian eigenvalues calculated at equilibrium.

The real parts were negative for $I_{\text{ext}} < 9.78$ μA/cm^2, so the equilibria were stable. Near $I_{\text{ext}} = 9.78$ μA/cm^2, a complex-conjugate pair crossed the imaginary axis and the equilibrium became unstable, leading to repetitive firing.

We performed two experiments:

1. Start the system at the equilibrium for $I_{\text{ext}} = 0$.
2. Start the system at the equilibrium for $I_{\text{ext}} = 9$ μA/cm^2 and then apply the same current.

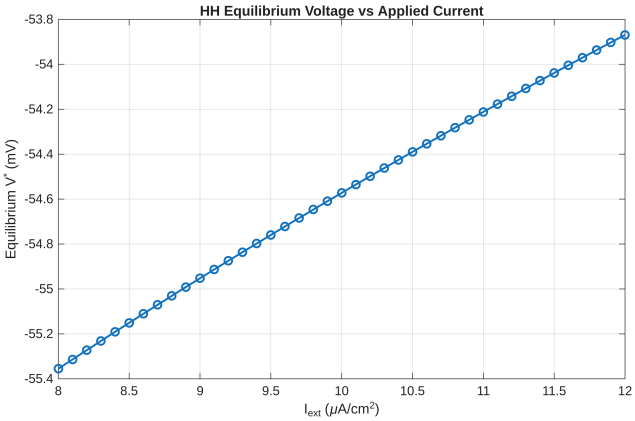


Figure 17: Q15: HH equilibrium voltage versus steady applied current. The equilibrium becomes progressively less negative as the current increases from 8 to 12 μA/cm^2.

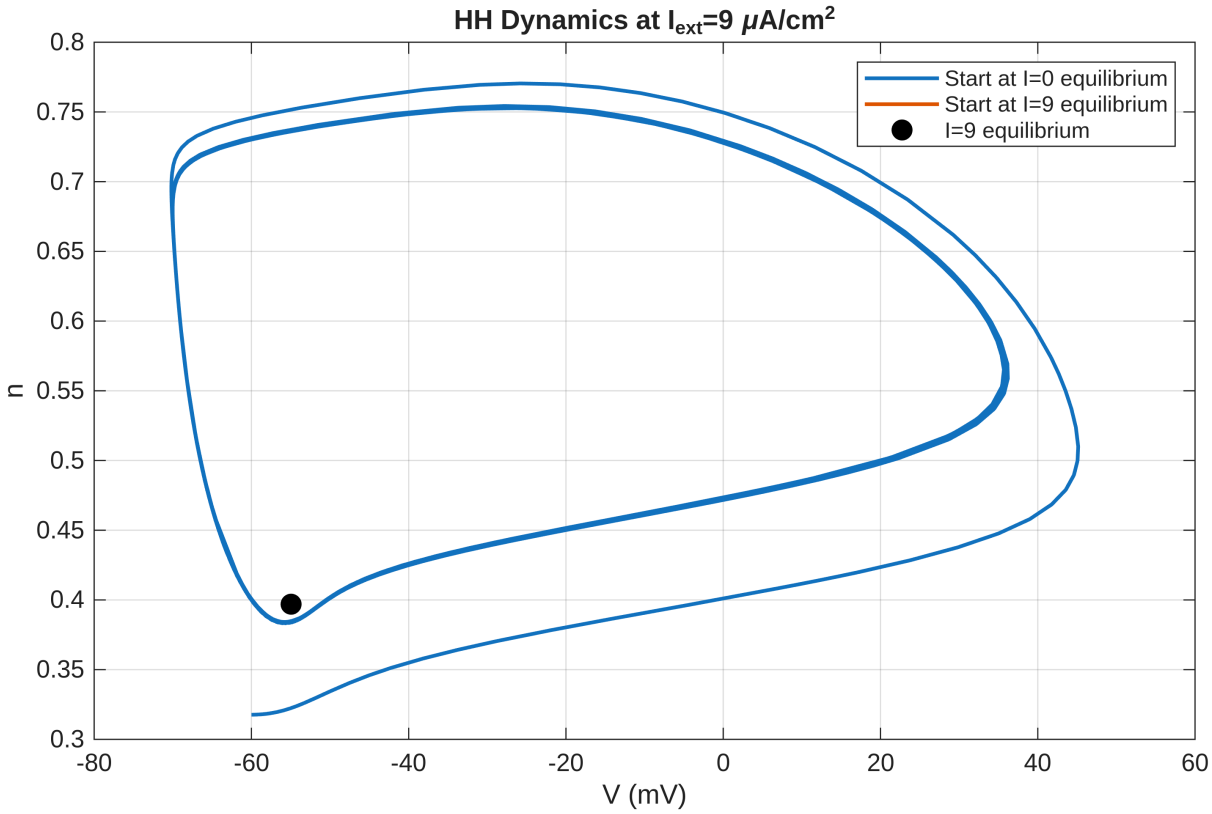


Figure 19: Q15: HH dynamics at $I_{\text{ext}} = 9\ \mu\text{A/cm}^2$ projected onto the V–n plane. Starting at the equilibrium for the applied current remains at the stable point, whereas starting from the zero-current equilibrium approaches the coexisting periodic orbit.

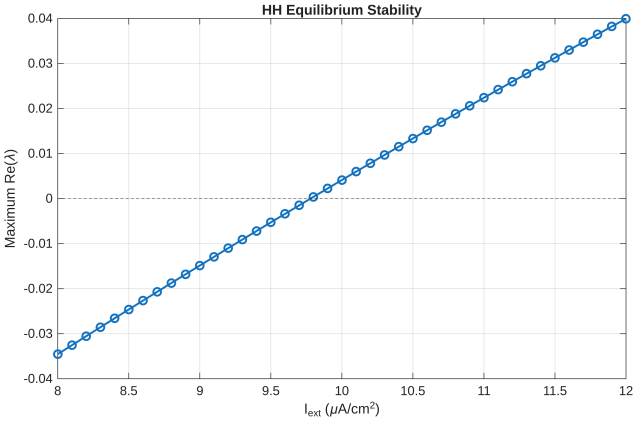


Figure 18: Q15: Maximum real part of the HH equilibrium eigenvalues for steady currents from 8 to 12 $\mu\text{A/cm}^2$. The zero crossing near 9.78 $\mu\text{A/cm}^2$ marks the loss of equilibrium stability.

Q16. Reduced V–n model

Assuming

$$m = m_\infty(V), \qquad h \approx 0.8 - n,$$

the HH model is reduced to the two variables V and n. The m gate is much faster than n and h, so m is assumed to instantly reach

$$m = m_\infty(V) = \frac{\alpha_m(V)}{\alpha_m(V) + \beta_m(V)}.$$

The variables n and h have similar slow time scales and are anticorrelated, so $h \approx 0.8 - n$.

The reduced HH model produces action potentials but differs from the original model in some key properties:

- **Peak voltage:** The reduced model has a higher peak voltage, approximately 50 mV compared with 45 mV in the original model. Since m is assumed instantaneous, it depolarizes more quickly and strongly.
- **Narrow action potential:** The reduced model has narrower spikes. Since there is no independent time dependence of m, it rises and falls faster.
- **Faster firing rate:** The reduced model fires faster. The approximation of m changes sodium activation, and the approximation of h changes sodium-inactivation timing, giving a different firing rate.

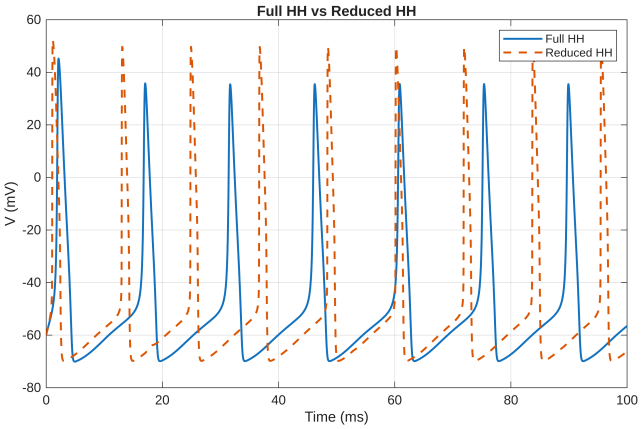


Figure 20: Q16: Voltage traces from the full and reduced HH models under the same applied current. The reduced model fires more rapidly and produces narrower spikes with a higher peak voltage.

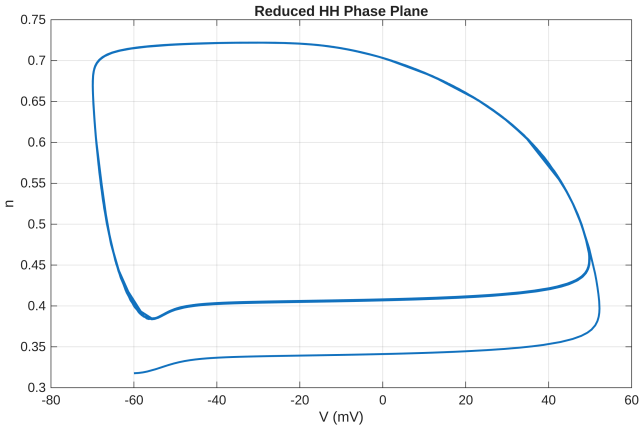


Figure 21: Q16: Periodic trajectory of the reduced HH model in the V–n phase plane.

Q17. Anode-break excitation

A hyperpolarizing current of $-3\ \mu\text{A/cm}^2$ was applied for 20 ms. During the pulse, the membrane potential became hyperpolarized, the potassium activation gate n decreased, and the sodium inactivation variable h increased. When the hyperpolarizing current was removed at $t = 20$ ms, the membrane rapidly depolarized and produced an action potential. This occurs because hyperpolarization increases sodium-channel availability and decreases potassium conductance.

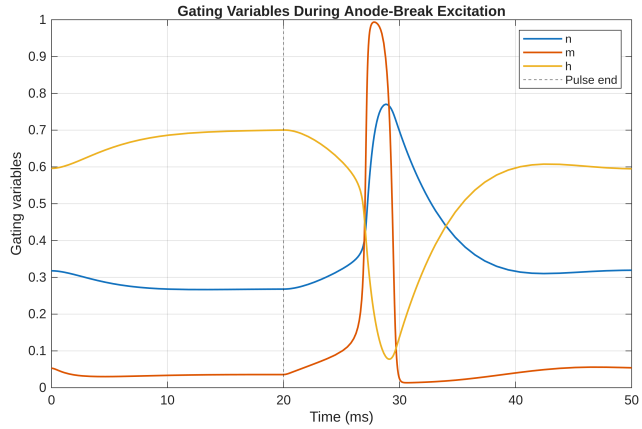


Figure 23: Q17: Evolution of the HH gating variables during anode-break excitation. During hyperpolarization, n decreases and h increases; after pulse removal, rapid activation of m initiates the spike.

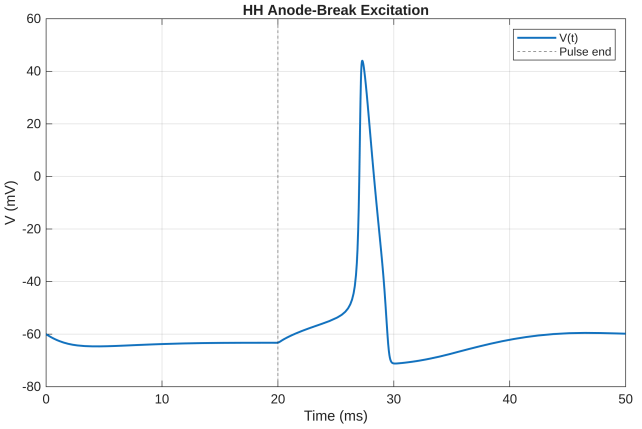


Figure 22: Q17: Anode-break excitation in the HH model. The dashed line marks the end of the 20 ms hyperpolarizing pulse; the action potential occurs after the stimulus is removed.

Q18. Reduced fast V–m subsystem

The HH model was reduced to the two variables V and m by fixing n and h. When n and h were fixed at their resting values, three equilibrium points were obtained, including the resting equilibrium at approximately

$$(V, m) \approx (-60,\ 0.0529),$$

with the other two near

$$(-57.3823, 0.0717) \quad \text{and} \quad (53.9187, 0.9992).$$

When n and h were fixed at their values at the end of the hyperpolarizing pulse,

$$n_{\text{end}} \approx 0.2680, \qquad h_{\text{end}} \approx 0.7002.$$

With these values, the resting equilibrium near -60 mV disappears, leaving only a stable high-voltage equilibrium near $(54.3488, 0.9992)$. Therefore, after removing

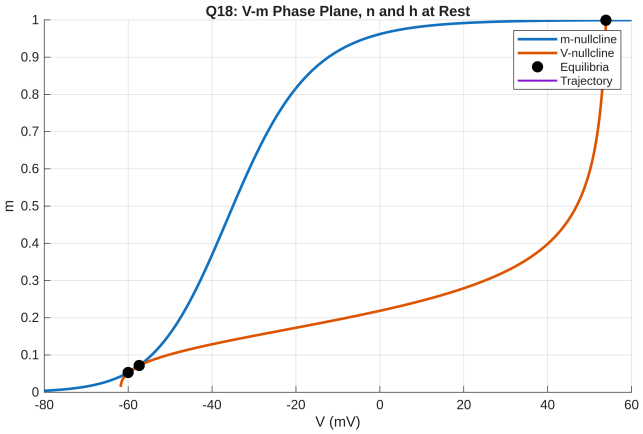


Figure 24: Q18, case 1: V–m phase plane with n and h fixed at their resting values. Three equilibria are present, including the resting equilibrium near -60 mV.

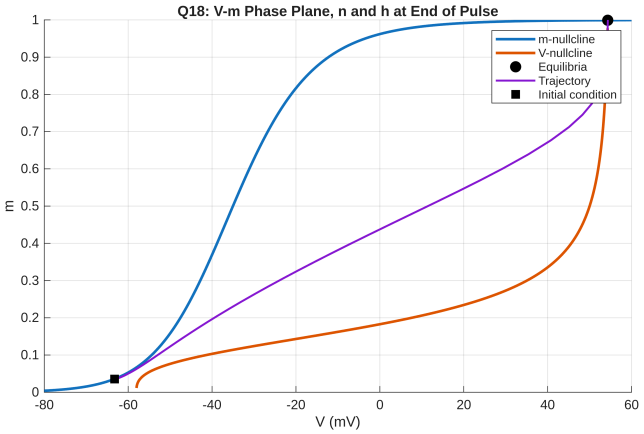


Figure 25: Q18, case 2: V–m phase plane with $n = 0.2680$ and $h = 0.7002$, their values at the end of the hyperpolarizing pulse. The low-voltage resting equilibrium has disappeared, leaving only the high-voltage equilibrium.

the hyperpolarizing current, the membrane is no longer attracted to its resting state. The fast m gate drives depolarization through increased sodium activation.